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Impact of task repetition schedules and emotions on L2 writing performance profiles using latent transition analysis

Mahmoud Abdi Tabari[a,1], Hansol Lee[b,*,2]

[a] *Department of English, University of Nevada, Reno, United States*

[b] *Department of English, Korea Military Academy, Seoul, Republic of Korea*

[*] Correspondence to: Department of English, Korea Military Academy, 574 Hwarang-ro, Nowon-gu, Seoul, Republic of Korea.

E-mail addresses: mabditabari@unr.edu (M. Abdi Tabari), hansol@kma.ac.kr (H. Lee).

[1] orcid.org/0000-0002-8022-5415

[2] orcid.org/0000-0002-6912-7128

Abstract

This study employed Latent Transition Analysis (LTA) to identify distinct L2 writing performance profiles based on CALF measures (Syntactic Complexity, Accuracy, Lexical Complexity, and Fluency) and to examine the influence of emotional states under varying task repetition schedules. Participants included 270 ESL undergraduates in a U.S. university writing course, and they were randomly assigned to long-spaced, short-spaced, or massed repetition schedules. Each completed a persuasive writing task, followed by a questionnaire measuring anxiety and enjoyment. Results indicated that learners initially displayed similar performance levels but ultimately formed two profiles: Lower Performance and Higher Performance. Those in the long-spaced group were more likely to belong to the Higher Performance Profile, characterized by higher CALF measures, particularly in syntactic complexity, accuracy, and lexical complexity. Among the predictors—three types of task repetition schedules (long-spaced, short-spaced, and massed repetitions) and two task-related emotions (enjoyment and anxiety)—perceived enjoyment emerged as the strongest predictor of being classified into the Higher Performance Profile. These findings emphasize the importance of utilizing a person-centered approach to explore individual differences in L2 writing trajectories and suggest that carefully designed task repetition schedules and attention to emotional states can enhance writing performance.

Keywords: Task repetition schedules; CALF profiles; L2 writing performance; Emotional states; TBLT; Latent Transition Analysis (LTA)

1. Introduction

Within the realm of Task-Based Language Teaching (TBLT), task repetition has emerged as a crucial variable for promoting language development. Repeating tasks provides learners with valuable opportunities to revisit and refine their linguistic output, supporting the gradual consolidation of knowledge and skills (DeKeyser, 2018). While the advantages of task repetition in enhancing oral performance have been extensively documented (see Bygate, 2018), comparatively less attention has been given to its effects on L2 writing, an area that presents distinct cognitive challenges beyond those encountered in extended oral tasks. Although both modalities require learners to manage linguistic complexity and achieve discourse-level accuracy and fluency (Johnson, 2024; Peng, 2024), L2 writing demands additional sustained attention to elements such as textual organization, syntactic complexity, and grammatical accuracy, without the benefit of real-time interlocutor feedback (Abdi Tabari, 2022). Given the cognitively demanding nature of L2 writing, which requires learners to engage in complex processes like planning, translating thoughts into written text, and self-monitoring, task repetition provides a crucial opportunity to support the development of writing skills over time. Repeating tasks allows learners to focus on refining specific aspects of their writing, such as organization, coherence, and lexical choice, while also offering the chance to internalize strategies for managing the cognitive load involved in writing (Manchón & Williams, 2016; Nückles et al., 2020). As learners engage in repeated task cycles, they progressively enhance their writing fluency and accuracy, which ultimately contributes to long-term skill development (Abdi Tabari et al., 2025).

One of the key challenges within TBLT research is identifying the optimal conditions under which task repetition can most effectively foster L2 writing development (Gilabert et al., 2016). Previous studies have focused largely on exact task repetition[3] and procedural repetition[4] (Tran & Saito, 2024), but little is known about how different *scheduling patterns*[5] of task repetition—whether tasks are repeated in massed, short-interval, or long-interval formats—affect learners' written performance. Insights from cognitive psychology, particularly research on the spacing effect (e.g., Cepeda et al., 2006), suggest that spaced practice[6] leads to better retention and performance compared to massed practice. While this effect has been mostly explored in L2 speaking tasks, the impact of repetition schedules on L2 writing performance, specifically on syntactic complexity, accuracy, lexical complexity, and fluency (CALF), remains underexplored (Abdi Tabari et al., 2024a).

Compounding this gap is the tendency of prior TBLT research to treat writing performance as a static, one-off outcome. Such an approach fails to capture the dynamic and evolving nature of learner development across repeated writing tasks (Nitta & Baba, 2014). Writing is a

recursive process that unfolds over time, and a single snapshot of performance does not adequately represent how learners progress (Kellogg & Whiteford, 2009). There is a growing need in TBLT to examine how learners' writing performance evolves through repeated exposure to tasks, particularly across different repetition schedules. This evolution can be better understood by tracking changes in performance over time, particularly through CALF measures.

While task repetition has been widely studied in relation to L2 writing performance, the joint influence of repetition schedules and emotional states (e.g., task anxiety and task enjoyment), remains underexplored (Abdi Tabari et al., 2025). Most prior studies focus on static performance outcomes, with limited attention to how learners' emotions evolve across different repetition intervals. Yet, emotional states can significantly shape attention, cognitive processing, and motivation during task execution (Fiedler & Beier, 2014). To address this gap, the present study investigates how task repetition timing interacts with emotional responses, offering a more dynamic perspective on L2 writing development and informing affect-sensitive task design.

To better understand this dynamic, the study employs Latent Transition Analysis (LTA) as an innovative framework to capture the developmental trajectory of L2 learners' writing performance across different task repetition schedules. LTA allows us to categorize learners into distinct performance profiles based on CALF measures and track how they transition between these profiles across multiple task repetitions. This approach moves beyond static measures of writing performance to provide a more process-oriented understanding of how task repetition and emotional states interact in shaping writing development in TBLT. Hence, the aims of this study are threefold: first, to categorize learners' writing performance profiles based on CALF measures; second, to examine how these performance profiles evolve as learners progress through the task repetition process; and third, to investigate how task repetition schedules (massed, short-interval, and long-interval) and learners' emotional states influence the likelihood of transitions between different writing performance profiles over time.

2. Review of Literature

2.1. The role of distributed practice in cognitive psychology and second language acquisition

Distributed practice is essential for effective learning, particularly in cognitive psychology and second language acquisition (SLA). It involves revisiting material at spaced intervals rather than relying on a single session. Research consistently shows that spaced learning—known as

the spacing effect—enhances retention more than massed practice or cramming (Carpenter, 2020; Kornell, 2009). This effect has been documented across domains including mathematics, verbal memory, and spelling, highlighting its broad applicability (Bahrick & Hall, 2005). Even when total study time is constant, spaced reviews yield better outcomes than cramming (Rohrer & Pashler, 2010). Despite strong evidence for its benefits, research on optimal spacing intervals remains inconclusive due to factors such as individual learning pace, task complexity, prior knowledge, and learner-related variables like motivation and fatigue (Abdi Tabari et al., 2025; Yavner, 2016). Theoretical models, such as study-phase retrieval theory (Benjamin & Tullis, 2010), explain why distributed practice is effective. This theory posits that spaced retrieval attempts reinforce learning, aligning with Bjork and Bjork's (2020) concept of desirable difficulty, where manageable challenges promote deeper learning. Unlike massed practice, which relies on immediate recall, spaced practice introduces delays that strengthen memory. Contextual variability further supports this effect, as shifting contexts between sessions provide richer retrieval cues (Tulving & Thomson, 1973). In contrast, massed practice may lead to habituation and reduced engagement (Dunlosky et al., 2013). Distributed practice also enhances higher-order skills beyond memorization (Benjamin & Tullis, 2010; Edmonds et al., 2021). Carolina et al. (2019) found that students using spaced reviews not only retained more but also transferred knowledge more effectively, supporting cognitive development in problem-solving and critical thinking.

In SLA, distributed practice—where learning sessions are spaced out over time—has proven particularly effective for vocabulary and grammar learning. Spaced retrieval, for example, can enhance long-term retention by up to 300 % compared to massed practice, in which learning occurs in a condensed timeframe (Nakata, 2015). However, the optimal spacing interval remains a matter of ongoing debate. While some studies advocate for longer intervals between repetitions (Bird, 2010), others report greater benefits from shorter gaps (Suzuki, 2017). Suzuki and Hanzawa (2021) argue that such discrepancies may stem from differences in task complexity and the cognitive demands imposed by the task itself.

A critical factor in this review is the distinction between declarative knowledge (e.g., vocabulary) and procedural knowledge (e.g., writing). Declarative knowledge may consolidate relatively quickly through limited exposures. In contrast, procedural knowledge requires repeated, distributed engagement for automatization, especially in complex, cognitively demanding tasks like writing (Saks et al., 2021; Suzuki & DeKeyser, 2017). Accordingly, a nuanced approach to distributed practice in SLA must consider not only the spacing interval but also the nature of the knowledge targeted, the task demands, and relevant cognitive and emotional factors.

3. Task repetition spacing in L2 writing

In the TBLT domain, task repetition (TR)—repeating the same or similar tasks—has been shown to enhance language development by offering learners opportunities for proceduralization and refinement of linguistic output. One important variable influencing the effectiveness of TR is spacing, or the time interval between repeated performances. Cognitive theories of learning, particularly deficient processing and distributed practice theories, suggest that spaced TR—in contrast to immediate or massed repetition—facilitates deeper processing and long-term retention by allowing forgetting to occur and necessitating retrieval, which strengthens memory traces (Cepeda et al., 2006; Nakata & Suzuki, 2019; Serrano, 2011).

In writing, where cognitive load is high and revision is central to improvement, these effects are particularly pronounced. Spaced TR provides learners with time for reflection and the opportunity to consolidate and reorganize knowledge between task performances. Immediate repetition may offer fluency gains due to reduced planning demands, but it often limits cognitive engagement and iterative development. In contrast, spaced repetition supports metacognitive and linguistic gains by allowing time for planning, revision, and strategic re-engagement with content.

Empirical evidence supports this distinction. Abrams (2019) found that learners in spaced TR conditions exhibited increased use of self-regulation strategies—such as planning and revising—compared to those in massed conditions. These learners paid more attention to structural elements and made more meaningful changes to their texts. Criado et al. (2022) similarly demonstrated that spacing helped learners reconnect with the communicative purpose of the task, resulting in written texts with greater syntactic complexity and lexical variety. These findings suggest that spaced TR not only promotes linguistic accuracy and sophistication but also fosters metacognitive awareness and productive engagement with writing as a process. As Delaney et al. (2010) observe, opportunities for reflection between tasks are essential for integrating linguistic forms into use, particularly in complex written production.

Despite these promising findings, much of the existing research examines spacing in binary terms—spaced versus massed—without systematically comparing different spacing intervals or exploring how spacing may influence distinct linguistic dimensions such as complexity, accuracy, fluency, and lexical variety. Additionally, few studies have considered the interplay between cognitive and affective dimensions in TR spacing. These gaps highlight the need for a more nuanced understanding of how spacing schedules shape writing development over time.

3.1. Empirical evidence on spacing intervals in task-based L2 writing

Recent research in task-based L2 writing has increasingly examined how varying spacing intervals between repeated writing tasks influence different linguistic outcomes. The literature primarily considers intervals ranging from one day to three weeks, but findings are mixed and sometimes contradictory.

Studies that explicitly manipulate and compare specific spacing intervals generally suggest differentiated effects depending on interval length. A one-week interval has emerged as a common moderate spacing condition. For example, Kim and Tracy-Ventura (2013)and Li and Dekeyser (2019) reported that spacing task repetitions by one week resulted in significant gains in syntactic complexity and accuracy over time. Similarly, Ajabshir and Poorebrahim (2021) found that learners repeating tasks with a one-week gap showed improvements in both accuracy and syntactic complexity, although fluency gains were limited. These findings highlight the potential of moderate spacing to enhance structural development in L2 writing.

In contrast, shorter intervals such as one day appear to primarily facilitate fluency gains. Huh et al. (2020), for instance, found that a one-day spacing between task repetitions improved learners' fluency but did not lead to significant gains in syntactic complexity or lexical variety. This suggests that shorter intervals may support automatization and smoother language retrieval, whereas longer intervals encourage more elaborated and reflective language use.

At the other end of the spectrum, longer intervals may reduce the effectiveness of repetition if they exceed an optimal threshold. Lu and Li (2023) observed that spacing repetitions over a three-week period led to declines in syntactic and lexical complexity, though accuracy remained stable. This implies that excessively long intervals might diminish retrieval benefits due to memory decay or reduced learner engagement.

Separately, several studies have examined the effects of repeated writing without explicitly manipulating or clearly specifying the spacing intervals. For example, Roothooft et al. (2022) conducted repeated writing tasks over approximately two weeks but did not systematically control the spacing between repetitions. They found that while repetition improved syntactic complexity, gains in fluency, lexical diversity, and accuracy were limited unless learners also received additional support such as modeling or feedback. Similarly, Khezrlou (2020) reported that task repetition alone, with intervals ranging from one to several days and without written corrective feedback, produced improvements in accuracy. These studies underscore that even when spacing is not the primary variable, task repetition can positively influence certain aspects of writing development.

Overall, this body of evidence points to the critical role of interval duration in shaping task repetition outcomes in L2 writing. Shorter intervals tend to benefit fluency, moderate intervals (around one week) support syntactic complexity and accuracy, and longer intervals may risk diminishing returns. Nonetheless, there remains no consensus on the optimal spacing schedule to balance gains across all dimensions of writing (i.e., CALF). Furthermore, the affective dimension of writing, such as learners' motivation and emotional engagement under different spacing conditions, remains underexplored. This highlights a clear need for future research to investigate how specific spacing intervals influence multiple dimensions of L2 writing over time.

4. Task emotions in TBLT: task anxiety and task enjoyment

While cognitive factors such as attention, working memory, and language aptitude have been extensively studied in TBLT (Fu & Li, 2019; Kormos & Trebits, 2012), emotional factors—particularly those related to task repetition schedules and L2 writing performance—have not received as much attention. However, recent research increasingly recognizes the role of emotions in shaping learners' engagement and task performance (Li & Dewaele, 2024). Learners' emotional responses, particularly task-specific emotions, play a critical role in modulating their attention, cognitive processing, and overall performance (Abdi Tabari et al., 2024b; Rahimi & Zhang, 2019). These emotions are not merely peripheral but central to the learning process (Havas & Matheson, 2013).

Task-specific anxiety, which arises from the situational demands of a task (e.g., novelty, cognitive load, or performance pressure), is distinct from trait anxiety, which reflects learners' general apprehension toward language learning (Jiang, 2021). Task-specific anxiety can enhance focus and engagement when learners perceive the task as manageable, but excessive anxiety may impair cognitive processing and hinder performance (Phung et al., 2021). This dual role of anxiety underscores the need for well-designed tasks that offer appropriate levels of challenge without overwhelming the learner. While task repetition may reduce anxiety over time as learners become more familiar with the task, it is unclear whether the intensity of task-specific anxiety increases or decreases across repeated tasks. Our previous study (Abdi Tabari et al., 2024b) provided preliminary evidence that task repetition influences fluctuations in task-specific emotions, but it focused exclusively on oral narrative tasks and used a cross-sectional design. The current study extends this line of inquiry by investigating how different spacing schedules of repeated writing tasks affect the dynamic interplay between task-specific anxiety and writing performance over time. By employing a longitudinal design and focusing on written output, this study offers a more comprehensive view of emotional trajectories in response to task repetition in L2 writing.

In contrast to anxiety, which can hinder performance under conditions of cognitive overload, task enjoyment is a positive affective response that has been shown to enhance motivation, engagement, and ultimately linguistic development. Enjoyment is often fostered by factors such as a sense of personal achievement, meaningful social interaction, task relevance, and constructive feedback (Dewaele & MacIntyre, 2014; Reeve, 2018). While anxiety tends to fluctuate with task demands, enjoyment typically increases over time as learners gain confidence, autonomy, and familiarity with task procedures (Abdi Tabari et al., 2024b; Dewaele et al., 2018). Despite its recognized importance, the interplay between task repetition and enjoyment remains underexplored. In our previous study (Abdi Tabari et al., 2024b), we neither manipulated the timing of task repetitions nor examined how enjoyment changes across successive writing sessions. The current study, employing a multi-session design over three weeks with varied spacing intervals, provides a more rigorous investigation of the temporal development of enjoyment during repeated L2 writing tasks, offering key insights into affective engagement dynamics across multiple exposures.

Importantly, emotions such as anxiety and enjoyment are not mutually exclusive. Recent findings suggest that these emotions can co-occur during complex tasks, where learners may experience both anxiety (due to the challenges posed by the task) and enjoyment (due to the satisfaction of overcoming these challenges) (Phung et al., 2021). This co-occurrence highlights the dynamic role emotions play in task engagement, with both positive and negative emotions driving learner motivation and engagement in different ways (Cuéllar & Oxford, 2018). Understanding these emotional dynamics is essential for examining how task repetition schedules and learners' emotional responses shape their cognitive processing and performance over time.

Although research on the relationship between task repetition and emotions such as enjoyment and anxiety is still emerging, recent studies have begun to address this gap. For instance, Abdi Tabari et al. (2025) examined the effects of massed and spaced task repetitions on L2 writing performance and task-related emotions. Their cross-sectional analysis revealed that while long-spaced repetitions can effectively reduce anxiety and enhance enjoyment, they may temporarily hinder fluency. In contrast, massed repetitions can lead to immediate improvements in performance but may negatively affect emotional responses. These findings highlight the need for continued research into how different repetition schedules influence both linguistic and affective dimensions of L2 writing.

Building on these findings, this study takes a broader perspective to examine how task-specific anxiety and enjoyment *change* across repeated writing tasks. Specifically, we explore how emotional responses to task repetition influence learners' transitions between different writing

performance profiles, such as shifts in CALF. By linking emotional fluctuations with language performance, this study aims to provide insights into how emotional engagement drives or hinders learners' progression across various writing tasks and repetition schedules.

5. The present study

Despite the existing research on task repetition and its effects on language learning, several significant gaps remain that this study aims to address. First, there is a limited understanding of how learners' writing performance profiles—measured by CALF—are categorized during task repetition schedules. Additionally, while some studies have highlighted the benefits of task repetition, there is a lack of in-depth analysis on how these performance profiles evolve over time. Finally, the interplay between the type of task repetition schedule and learners' emotional states has not been thoroughly explored, particularly regarding how these factors may influence transitions between different writing performance profiles. To address these gaps, this study seeks to answer the following research questions:

RQ 1: How are learners' writing performance profiles, based on CALF measures, categorized in Task Repetition activities?

RQ 2: How do learners' writing performance profiles evolve throughout the Task Repetition activities?

RQ 3: How do the predictors, such as three types of Task Repetition schedules (long-spaced, short-spaced, and massed repetitions) and two task-related emotions (enjoyment and anxiety), influence the likelihood of transitioning into different writing performance profiles?

To address these research questions while capturing the complexity and diversity of individual learner experiences, we applied Latent Transition Analysis (LTA) from a person-centered perspective. Unlike traditional variable-centered approaches—such as MANOVA or MANCOVA—that focus on group-level averages and assume homogeneity within groups, a person-centered approach highlights the identification of distinct learner profiles and their dynamic trajectories over time (Crowther et al., 2021; Staples & Biber, 2015). LTA is particularly well-suited for the purpose of the current study, as this approach identifies latent subgroups within the data and models how individuals transition between these subgroups over time (Lee et al., 2023; Demaray et al., 2021). This dynamic modeling capability offers a nuanced understanding of how learners' profiles evolve in response to instructional practices, such as task repetition. While LTA has limitations, including its reliance on large sample sizes and the complexity of model construction, its ability to uncover time-sensitive patterns in

learner trajectories justifies its use for reanalyzing existing data. By leveraging LTA, this study aims to explore diverse learner profiles and gain deeper insights into the impact of task repetition on writing performance.

6. Methods

6.1. Participants and learning context

This study was conducted with 270 ESL students enrolled in several sections of a first-year undergraduate ESL writing course at a large U.S. university. The participants were selected through convenience sampling and included 120 men and 150 women, aged between 18 and 22 (M = 20.5). They represented a range of first language (L1) backgrounds: Arabic (15 %), Dutch (10 %), Japanese (10 %), Chinese (15 %), French (10 %), Spanish (15 %), Korean (10 %), Indian languages (10 %), and Portuguese (5 %). Most had resided in the U.S. for less than 18 months (85 %), while the remainder had been in the country for 3–5 years. Their initial writing proficiency was determined using University Writing Placement Scores (range: 7–8 out of 9). On average, participants had completed 2.5 ESL writing courses prior to this one, with a range from zero to five courses. All participants had received approximately 30 h of classroom instruction before the study began. To ensure comparable proficiency across groups, the IELTS Task Version 2 writing test was administered at baseline (Week 5), with scores ranging from 6.5 to 7.5, corresponding to the Advanced High level on the ACTFL scale.

7. Materials

7.1. Writing task

The study used a written persuasive task to evaluate first-year ESL students' writing skills for several reasons. This type of writing is essential in academic settings, as it helps students learn to construct clear arguments and express their viewpoints effectively. Additionally, the task promotes critical thinking by requiring students to analyze different sides of an issue and evaluate evidence. It also aligns with the goals of the ESL writing curriculum in many universities in the U.S., focusing on improving argumentative writing and organizing ideas logically. Furthermore, engaging with relevant topics makes the task more meaningful and motivating for students (Abdi Tabari, 2023). The writing prompt was provided as follows:

The university is thinking about increasing funding for its study abroad programs. As a student representative, your job is to write a persuasive essay for the university administration. In your essay, discuss the pros and cons of studying abroad, using examples and evidence to support

your points. Finish by recommending whether or not the university should move forward with this plan.

7.2. Emotion questionnaires

To capture the nuances of anxiety and enjoyment experienced by students in their immediate and contextual settings over time, we adopted Jiang's (2021) scales for task anxiety and task enjoyment, specifically tailored for the writing task. The task enjoyment scale consisted of five items, each rated on a 6-point Likert scale, where participants indicated their level of enjoyment ranging from 1 (not at all) to 6 (very much). For instance, an example item is "I enjoyed doing the task," which encourages students to reflect on their overall satisfaction with the writing experience. Conversely, the task anxiety scale comprised six items that measured the feelings of apprehension associated with the writing task. This scale also utilized a 6-point Likert scale, with responses ranging from 1 (strongly disagree) to 6 (strongly agree). An illustrative item from this scale is, "When completing the writing task, I felt my heart pounding," prompting participants to consider their physiological responses during the task. The reliability of these scales was rigorously assessed, revealing Cronbach's alpha coefficients of .96 for the task anxiety scale and .95 for the task enjoyment scale. These high coefficients indicate strong internal consistency, confirming the scales' effectiveness in measuring the emotional responses of students during the writing task.

8. Procedure

Initially, we contacted the Director of the Writing Program to provide a clear explanation of the study's aims and procedures. Following this, we reached out to the instructors of first-year ESL writing courses to seek their support in coordinating the study's logistics. The first author conducted the writing tasks and administered the post-task emotion questionnaires during scheduled class visits, ensuring that all participants clearly understood the study's objectives, procedures, and their rights concerning voluntary participation and confidentiality.

To ensure that all participants had similar levels of English writing proficiency, we administered the IELTS Task Version 2 writing test in Week 5. Upon completion of this pre-test, 270 participants were randomly assigned to one of three equally sized groups (n = 90 per group), each following a specific task repetition schedule. This study is a secondary analysis of a dataset originally collected as part of a broader investigation (Abdi Tabari et al., 2025) that explored the immediate effects of task repetition schedules on L2 writing using a variable-centered approach. While that earlier work focused on overall changes in mean-level CALF scores across different repetition schedules, the current study takes a person-centered approach and introduces a novel focus on learners' dynamic performance transitions over time.

Specifically, we employed Latent Transition Analysis (LTA) to examine how learners moved between discrete writing performance profiles, with particular attention to the interplay between task schedules and emotional variables such as task enjoyment and anxiety. Thus, the present work offers a new analytic lens and addresses different theoretical and pedagogical questions than those pursued in the prior publication.

The three task repetition conditions were as follows:

1. *Long-spaced repetition*: Participants performed the persuasive writing task once per week over three consecutive weeks (Weeks 6–8).
2. *Short-spaced repetition*: Participants completed the task every two days within one week (Monday, Wednesday, and Friday of Week 6).
3. *Massed repetition*: Participants completed all three iterations consecutively within a single class session.

Each task iteration lasted 35 min, a duration determined by pilot study results as optimal for task completion among participants with comparable English proficiency. Table 1 below summarizes the scheduling of the task iterations across groups.

After each iteration, participants completed emotion questionnaires assessing task enjoyment and anxiety using hard-copy forms. The essays produced during each iteration (initial, interim, and final assessments) were analyzed for detailed CALF metrics. The essays averaged 367 words in length (SD = 69), with individual essay lengths ranging from 309 to 458 words.

8.1. Analysis of L2 written production using the CALF metrics

To analyze L2 written production, we applied the CALF framework, which examines key aspects of writing through syntactic complexity, accuracy, lexical complexity, and fluency. Each construct was assessed using well-established metrics, ensuring comprehensive and reliable analysis aligned with prior research on L2 writing. Below is a description of the metrics used and the rationale for their selection.

Table 1
Study Design Overview.

	Monday	**Wednesday**	**Friday**
Week 6	Massed IT1-IT2-IT3 Short-spaced IT1 Long-spaced IT1	Short-spaced IT2	Short-spaced IT3
Week 7	Long-spaced IT2		
Week 8	Long-spaced IT3		

Note. IT stands for iteration, so IT1, IT2, and IT3 are for the first iteration, second iteration, and third iteration of the task. Adapted from Abdi Tabari et al. (2025).

1. **Syntactic complexity:** Syntactic complexity reflects the structural sophistication of learner output. We used the L2 Syntactic Complexity Analyzer (L2SCA) developed by Lu (2010) to automatically calculate three metrics, such as MLT, DC/T, and CN/T, ensuring accuracy and consistency across samples. Bivariate correlations among the complexity metrics were moderate ($r < .70$), confirming that each captures a distinct aspect of syntactic structure. The definitions of the three metrics are as follows:
    - **Mean Length of T-unit (MLT):** This metric gauges general syntactic complexity by dividing the total number of words by the total number of T-units. A T-unit is defined as a main clause along with any attached subordinate clauses or non-clausal elements (Hunt, 1970). A higher MLT indicates more elaborated sentence structures.
    - **Ratio of Dependent Clauses to T-units (DC/T):** This metric captures subordination complexity, revealing the extent to which learners use dependent clauses to expand their ideas (Norris & Ortega, 2009).
    - **Ratio of Complex Nominals per T-unit (CN/T):** This metric measures phrasal complexity, focusing on how frequently complex noun phrases appear. This feature, typical in academic writing, reflects advanced control over nominal structures (Lu, 2010).
2. **Accuracy:** Accuracy was gauged as the proportion of error-free clauses in each text. Errors were categorized as syntactic, morphological, or lexical, with non-grammatical errors (e.g., punctuation) excluded following Wigglesworth and Storch (2009), since they do not impair meaning. To ensure reliability, 50 % of the texts were independently coded by an expert colleague. A high inter-coder agreement was achieved ($r = .94$) for error-free clauses and ($r = .96$) for correct verb forms, confirming the reliability of the coding process. This focus on local grammatical control provides insights into the learners' ability to produce structurally accurate output.
3. **Lexical Complexity:** We analyzed lexical complexity from two perspectives: diversity and sophistication.
    - **Lexical diversity:** We employed the Measure of Textual Lexical Diversity (MTLD), which captures the average length of word strings that maintain a set level of lexical variation (Abdi Tabari et al., 2023). MTLD is text-length independent, ensuring reliable comparisons across samples of varying lengths (McCarthy & Jarvis, 2010).
    - **Lexical sophistication:** This sub-construct was measured using the log frequency of content words from the CELEX database (WRDFRQmc). This metric, proposed by Kyle and Crossley (2015), is sensitive to subtle improvements in

learners' ability to use less frequent or more advanced words, reflecting vocabulary development in their writing.

4. **Fluency:** L2 writing fluency was operationalized as words per minute (W/M), which captures the rate of text production. This metric, based on Wolfe-Quintero et al. (1998), reflects the learners' ability to produce coherent text efficiently. As noted by Kellogg (1996), production rate serves as a valid product-based indicator of fluency, as it reflects both cognitive processing during writing and the learner's ability to sustain focus and generate content over time.

8.2. Data analysis

The current study reanalyzes the previously collected dataset in Abdi Tabari et al. (2025) using Latent Transition Analysis (LTA) to explore the dynamic changes in latent profiles of writing performance after repetitions using Stata version 16 (StataCorp, 2019). The LTA was conducted using a stepwise approach to identify latent profiles and examine profile transitions between the initial assessment and final assessment. The overall analysis plan was aimed at maximizing statistical power and enabling a holistic view of profile transitions across the entire sample before exploring the effect of interventions (i.e., different task repetition schedules) and emotions (i.e., task anxiety and enjoyment) following practical guidelines presented by Lee et al. (2023).

Step 1: identification of latent profiles

Initially, LTA was applied to the entire sample without splitting it by intervention type. This decision was made to preserve statistical power and prevent premature subgroup analyses that could diminish sample size and obscure potential patterns. Latent Profile Analysis (LPA) was conducted at each time point to identify distinct latent profiles based on the participants' writing performance as measured by the CALF metrics. Since the eight CALF metrics were on different scales, they were first standardized (mean = 0, standard deviation = 1) to ensure comparability and address scaling issues before being used in the LPA analyses. The results of LPA analyses did not automatically indicate the optimal number of latent profiles but provided values for five criteria to guide the determination of the best-fitting solution. The selection of the optimal number of profiles was based on multiple considerations, including profile size (ensuring profiles represented at least 5 % of the sample to avoid overfitting), the Bayesian Information Criterion (BIC; lower values indicating better fit), Akaike Information Criterion (AIC; lower values indicating better fit), log-likelihood (LL; higher values indicating better fit), Lo-Mendell-Rubin (LMR) likelihood ratio test (<.05), and entropy (>.80). Additionally, theoretical interpretability was a key factor, prioritizing the identification of meaningful and distinguishable profiles.

Step 2: transition analysis

Once the latent profiles were established for the initial assessment and final assessment, a transition matrix was constructed to examine how individuals moved between these profiles. This matrix provided insights into the stability of the profiles and the frequency of transitions. Profile stability and movement patterns were analyzed for the entire sample before introducing any covariates or predictors.

It should be noted that profiles at the interim assessment were not analyzed, and the transition analysis was conducted solely between the initial assessment and final assessment. This decision was made to focus on the primary time points of interest—before and after the repetitions—where meaningful shifts in learner profiles were most likely to occur. While including profiles at the interim assessment (i.e., between the first and second repetitions) could have provided a more dynamic assessment of learners' writing performance and emotional states across different task repetition schedules, it would have also introduced considerable complexity in interpreting fluctuating changes across three time points and may have emphasized transient variations rather than the broader developmental patterns that are central to understanding the impact of the intervention. By concentrating on the initial and final assessments, the analysis can remain focused, interpretable, and aligned with the study's aim of investigating pre- and post-repetition changes, while avoiding the potential for overcomplication due to short-term fluctuations. Given the space limitations for the article, prioritizing a clear and concise analysis was considered necessary.

Step 3: intervention and emotional covariates

Following the identification of profiles and transitions, the intervention type was added as a predictor of both profile membership and transition probabilities. A series of logistic regression models were employed to test whether different interventions influenced profile transitions over time. Additionally, emotional variables were tested to assess their impact on transitions between latent profiles. Moreover, any remaining independent variables, such as learner factors like placement test scores, age, and gender, were included in the logistic regression models to investigate any significant impact on profile transition patterns. When any of these variables showed statistical significance, we further investigated interactions between them. For example, if intervention type and emotions were found to be significant, we would further explore interactions between intervention type and emotions to determine whether emotional patterns moderated the effects of interventions on profile changes.

Given the complexity of combining person- and variable-centered approaches, it is important to clarify that the follow-up variable-centered analysis (e.g., logistic regression) was conducted post hoc to complement, not contradict, the person-centered findings. While the

person-centered approach was used to identify distinct learner profiles and account for individual variability, the subsequent variable-centered analysis was intended to explore how external factors (e.g., task repetition schedule, emotional states) were associated with transitions between these profiles. This combined approach aimed to provide a more comprehensive examination of profile dynamics, interventions, and emotions without prematurely dividing the sample, ensuring robust statistical analysis and more meaningful insights into the patterns of change across the sample.

9. Results

RQ 1: How are learners' writing performance profiles, based on CALF measures, categorized in task repetition schedules?

To answer the first research question, we conducted two LPAs for the collected CALF results at the initial assessment and final assessment. First, as shown in Table 2, the results of the first LPA indicated that the participants' writing performance profiles were best described when categorized into two profiles. In terms of the profile size, solutions with more than four profiles could not be accepted due to the overfitting issue, meaning that solutions with two, three, and four profiles were considered. When it comes to the BIC, the solution with three profiles had the lowest value followed by the one with two profiles with a slim margin, leaving the one with four profiles out from the finalists. The AIC and LL indices as well as the LMR test results all indicated that solutions with more profiles tended to have better fits, favoring the solution with three profiles over the one with two profiles. However, only the solution with two profiles satisfied the entropy index with a value of.94, while the rest, including the solution with three profiles, had way too low values. Based on these indices, we selected the solution with two profiles as the best model for learners' writing performance profiles at the start of the repetition (i.e., initial assessment).

According to the two-profile solution, illustrated in Fig. 1 (see Table 3 for descriptive statistics for each profile), the participants were categorized into two groups, which we labeled (1) Typical Profile and (2) Unexplained Profile, based on their writing performance measured by the CALF metrics. The Typical Profile, comprising 93 % of the sample, had standardized means close to zero across all eight CALF metrics, indicating that this group had scores closely aligned with the overall average values for the entire sample. This alignment reflects their typical or average performance across these metrics. In contrast, the Unexplained Profile, representing only 7 % of the sample, exhibited relatively higher standardized means across seven of the eight metrics, particularly in accuracy, but had a notably lower mean for MTLD (lexical complexity). This distinctive pattern is challenging to interpret, as it does not readily align with existing theoretical frameworks or patterns commonly observed in similar research.

While it is difficult to exhaustively consider every possible explanation, the lack of an obvious theoretical basis for this combination of characteristics underscores the “unexplained” nature of this profile. The small size of this group suggests it may represent an outlier or a distinctive pattern that merits further exploration.

Second, in analyzing learners’ writing performance profiles after the repetitions (final assessment) to address the remaining part of the first research question, the results of the second LPA, shown in Table 4, indicated that the solution with three profiles was the best model. Solutions with more than three profiles were excluded due to overfitting, leaving only the two- and three-profile solutions for consideration. Regarding the BIC, the two-profile solution had a slightly lower value than the three-profile solution, indicating the former may have a better fit. However, the AIC and LL values indicated the opposite, favoring the three-profile solution. While the LMR test results indicated that both the two- and three-profile solutions were viable, this index did not decisively favor one over the other. Lastly, the three-profile solution had a higher entropy value (0.72) than the two-profile solution, which had a considerably lower value (0.58), although the entropy values for both solutions were below the recommended cutoff of.80.

Table 2

Fit Indices for Writing Performance Profiles at the Initial Assessment.

# of Profiles	Profile size	BIC	AIC	LL	Entropy	LMR test
2	**252, 18**	**6164.28**	**6074.32**	**−3012.16**	**.94**	**< .001**
3	149, 104, 17	6162.53	6040.18	−2986.09	.67	< .001
4	48, 95, 109, 18	6174.35	6019.62	−2966.81	.66	< .01
5	14, 139, 9, 90, 18	6193.73	6006.61	−2951.30	.78	< .05

Note. The selected best model is in bold. Underlined values for the profile size indicate profiles consisting of less than or equal to 5 % (i.e., 14) of the total sample. BIC = Bayesian information criterion; AIC = Akaike information criterion; LL = log-likelihood; and LMR test = Lo-Mendell-Rubin adjusted likelihood ratio test.

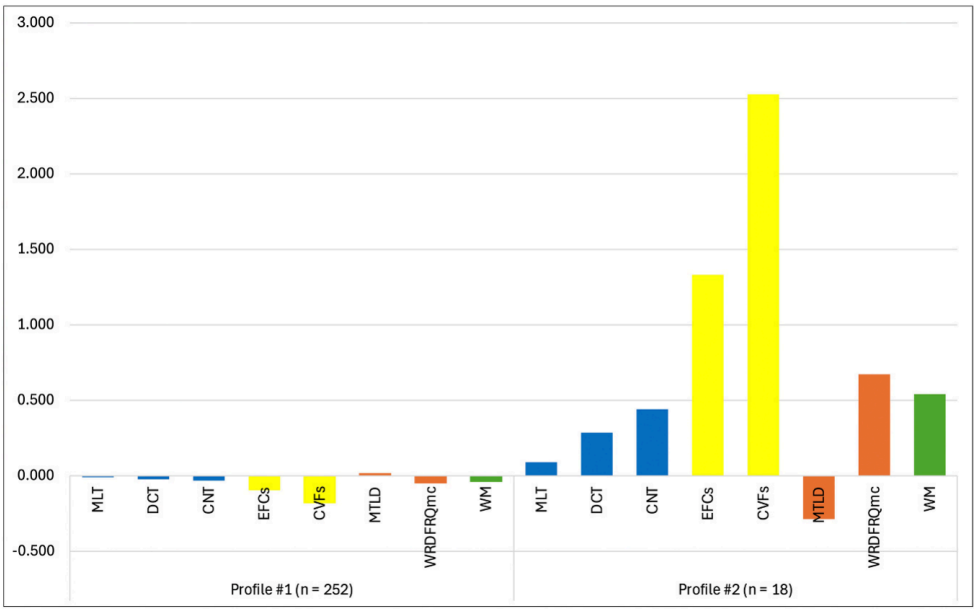


Fig. 1. *Writing Performance Profiles (Two Profiles) at the Start of the Intervention. Note*. MLT = the mean length of T-unit; DC/T = the ratio of dependent clauses to T-units; CN/T = the ratio of complex nominals per T-unit; EFCs = error-free clauses; CVFs: correct verb forms; MTLD = the measure of textual lexical diversity; WRDFRQmc = the log frequency of content words; and W/M = words per minute.

Table 3

Descriptive Statistics for Writing Performance Profiles at the Initial Assessment.

Profile	**Measurement**	**Standardized**			
		Mean	**SD**	**Min**	**Max**
Profile #1	MLT	−0.01	1.01	−2.72	4.78
(n = 252)	DCT	−0.02	0.98	−2.93	2.49
	CNT	−0.03	1.00	−4.01	3.91
	EFCs	−0.10	0.91	−3.82	3.75
	CVFs	−0.18	0.74	−3.21	1.70
	MTLD	0.02	0.98	−3.31	3.58
	WRDFRQmc	−0.05	0.98	−3.48	2.07
	WM	−0.04	0.94	−2.78	4.36
Profile #2	MLT	0.09	0.81	−1.50	1.34
(n = 18)	DCT	0.29	1.21	−2.79	1.94
	CNT	0.44	0.95	−1.62	1.50
	EFCs	1.33	1.29	−0.43	3.39
	CVFs	2.53	0.74	1.04	3.97
	MTLD	−0.28	1.20	−1.87	2.36
	WRDFRQmc	0.67	1.00	−1.05	2.01
	WM	0.54	1.58	−1.39	4.43

Note. The acronyms are defined in Fig. 1.

Table 4
Fit Indices for Writing Performance Profiles at the Final Assessment.

# of Profiles	Profile size	BIC	AIC	LL	Entropy	LMR test
2	142, 128	6150.95	6060.99	−3005.49	0.58	< .001
3	**133, 110, 27**	**6151.20**	**6028.85**	**−2980.42**	**0.72**	**< .001**
4	121, 12, 115, 22	6130.11	5975.38	−2944.69	0.77	< .001
5	108, 14, 106, 16, 26	6143.54	5956.427	−2926.21	0.79	< .01

Note. The selected best model is in bold. Underlined values for the profile size indicate profiles consisting of less than or equal to 5 % (i.e., 14) of the total sample. BIC = Bayesian information criterion; AIC = Akaike information criterion; LL = log-likelihood; and LMR test = Lo-Mendell-Rubin adjusted likelihood ratio test.

In addition to the indication of these indices for the determination for the optimal number of profiles, a detailed examination of the three-profile solution revealed seemingly plausible patterns in participants' writing quality development, leading us to select the three-profile solution as the most appropriate model for learners' writing performance profiles after the repetitions (final assessment). In the three-profile solution, shown in Fig. 2 (see Table 5 for descriptive statistics for each profile), we could label these profiles (1) Lower Performance Profile, (2) Higher Performance Profile, and (3) Unexplained Profile, based on their writing performance as measured by the CALF metrics. Considering the standardized CALF metrics enabled comparisons among participants across different metrics, the differences between profiles are interpreted as relative rather than absolute. Thus, participants in the Lower Performance Profile tended to score lower compared to their peers in the Higher Performance Profile. Importantly, this does not imply that those in the Lower Performance Profile experienced deterioration or that those in the Higher Performance Profile achieved exponential improvement. Instead, the differences reflect relative performance levels, with the Lower Performance Profile achieving relatively lower outcomes and the Higher Performance Profile achieving relatively higher outcomes, as the labels suggest. Therefore, the three-profile solution highlighted distinct and plausible patterns in participants' writing quality development.

Specifically, the Lower Performance Profile, accounting for 49 % of the sample, had standardized means around −0.5 for six of the eight metrics, indicating relatively lower performance compared to the group average. The Higher Performance Profile, comprising 41 % of the sample, exhibited standardized means of approximately +0.5 for six of the eight metrics, signifying relatively stronger performance. Notably, both profiles showed minimal differences in complexity (DCT) and fluency (WM), which warrants further discussion. Beyond these two groups, 10 % of the sample formed a third group, labeled (3) Unexplained Profile. This group displayed higher standardized means for three specific metrics, particularly

accuracy and WRDFRQmc, but their pattern did not align well with existing theoretical frameworks. Similar to its counterpart in the two-profile solution at the initial assessment, this group appears to represent outliers or a distinct subgroup requiring additional investigation.

Upon closely examining the three-profile solution across different TR schedules at the final assessment (see Table 6), we found that the Lower Performance Profile comprised 48 % of the massed group, 37 % from the short-spaced group, and 14 % from the long-spaced group. In contrast, the Higher Performance Profile displayed the opposite proportions, with 58 % from the long-spaced group, 31 % from the short-spaced group, and 11 % from the massed group. As the smallest profile, the Unexplained Profile consisted of 48 % of the massed group, with 26 % of the long-spaced group and another 26 % of the short-spaced group. Unlike the Typical Profile at the beginning of the repetition (initial assessment), where no significant differences in writing performance were observed, the results at the end of the repetition (final assessment) revealed notable variations across the different TR schedules.

RQ 2: How do learners' writing performance profiles evolve throughout the task repetition schedules?

Fig. 3 illustrates how learners' writing performance profiles changed throughout the task repetition activities from the initial assessment to the final assessment. First, the results show that the majority of the sample (i.e., Typical Profile, 93 %) was relatively homogeneous in terms of their writing performance based on the CALF metrics at the start of the intervention. After the intervention, in which learners were randomly assigned to three different groups with varying task repetition schedules, they were primarily divided into two contrasting profiles: Lower Performance (51 %) and Higher Performance (41 %). In addition to this main shift, we found that 7.5 % of those in the Typical Profile at the initial assessment transitioned to the Unexplained Profile after the intervention. Second, regarding those in the Unexplained Profile at the initial assessment, the largest portion (44 %) maintained their profile status throughout, while 22 % shifted to the Lower Performance Profile and 33 % moved to the Higher Performance Profile at the final assessment.

It is important to note that there is no direct equivalence between the profiles at Times 1 and 3 in this scenario, so we cannot infer any underlying meanings across the two time points. For example, if the profiles at both time points had the same labels, we could interpret changes more meaningfully—such as learners maintaining their profile (e.g., Proficient Profile at the initial assessment → Proficient Profile at the final assessment), improving their achievement (e.g., Lower Performance Profile at the initial assessment → Higher Performance Profile at the final assessment), or becoming less motivated (e.g., Motivated Profile at the initial assessment → Less Motivated Profile at the final assessment). However, in this case, since the profiles at

the initial assessment and the final assessment are not comparable (i.e., they have different labels), except for the Unexplained Profile, we focused on how learners were categorized into different profiles at the final assessment 3, regardless of their original profile at the initial assessment.

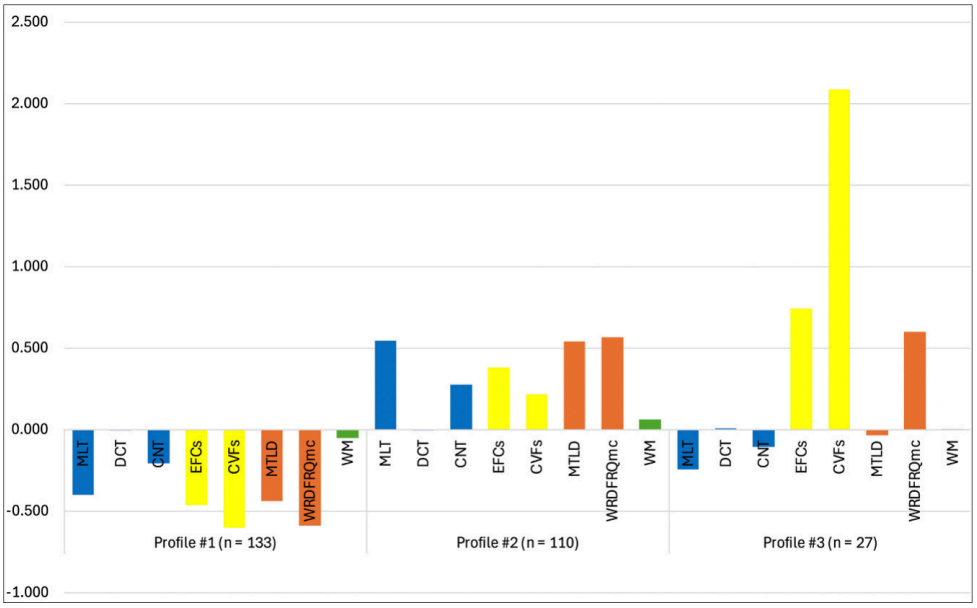


Fig. 2. *Writing Performance Profiles (Three Profiles) at Final Assessment. Note*. The acronyms are defined in Fig. 1.

Table 5
Descriptive Statistics for Writing Performance Profiles at the Final Assessment.

Profile	Measurement	Standardized			
		Mean	SD	Min	Max
Profile #1	MLT	−0.40	0.60	−1.82	1.41
(n = 133)	DCT	0.00	0.86	−2.93	2.40
	CNT	−0.21	0.80	−2.30	2.73
	EFCs	−0.47	0.88	−3.25	2.60
	CVFs	−0.60	0.63	−2.06	1.41
	MTLD	−0.44	0.83	−2.75	2.43
	WRDFRQmc	−0.59	0.82	−4.78	1.86
	WM	−0.05	1.05	−2.09	7.28
Profile #2	MLT	0.55	1.11	−1.61	4.88
(n = 110)	DCT	0.00	1.18	−3.33	2.48
	CNT	0.28	1.14	−2.30	5.60
	EFCs	0.38	0.81	−1.53	3.11
	CVFs	0.22	0.59	−1.10	1.50
	MTLD	0.54	0.87	−1.07	2.61
	WRDFRQmc	0.57	0.81	−0.98	2.26
	WM	0.06	0.92	−1.46	3.70
Profile #3	MLT	−0.24	1.09	−2.69	1.83

Profile	Measurement	Standardized			
		Mean	SD	Min	Max
(n = 27)	DCT	0.01	0.86	−1.36	1.78
	CNT	−0.11	1.04	−2.27	1.49
	EFCs	0.74	1.16	−1.10	2.85
	CVFs	2.09	0.55	1.15	3.23
	MTLD	−0.04	1.19	−2.26	2.09
	WRDFRQmc	0.60	0.84	−1.14	2.10
	WM	0.00	1.06	−1.54	2.86

Note. The acronyms are defined in Fig. 1.

Table 6

Distribution of Participants across TR Schedules by Writing Performance Profile at the Final Assessment.

TR Schedule	Lower Performance Profile (*n* = 133)	Higher Performance Profile (*n* = 110)	Unexplained (*n* = 27)
Long-spaced	19 (14 %)	64 (58 %)	7 (26 %)
Short-spaced	49 (37 %)	34 (31 %)	7 (26 %)
Massed	65 (49 %)	12 (11 %)	13 (48 %)

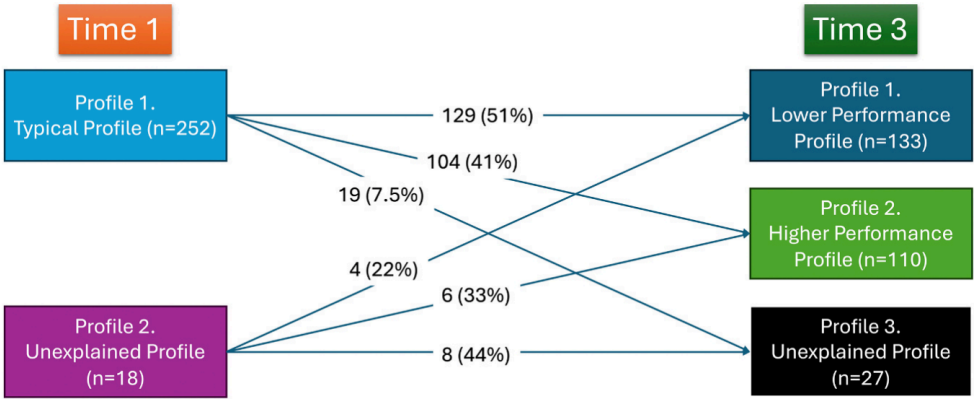


Fig. 3. Writing Performance Profile Transition Patterns Throughout the Intervention.

RQ 3: How do the types of Task Repetition schedule and learners' emotional states influence the likelihood of transitioning into different writing performance profiles?

This leads us to the third research question: What factors caused these changes in learners' writing performance profiles to occur? To address this, we examined whether task repetition types (e.g., long-spaced, short-spaced, massed), task-related emotions (e.g., enjoyment, anxiety; see Table 7 for their descriptive statistics), and other learner factors, such as placement test scores, age, and gender, significantly influenced the various profile transition patterns.

First, regarding the transition to the “Lower Performance Profile,” Table 8 presents the results of logistic regression models, showing the odds ratios for transitioning to this profile based on task repetition schedules and task emotions. For the task repetition schedules, a categorical independent variable, the results represent comparisons between the reference group (i.e., Short-spaced group, in this case) and other groups, in terms of the odds of being categorized into the Lower Performance Profile. The results indicated that the odds of being categorized into this profile were 77 % lower (i.e., $e^{-1.49} \approx 0.23$) for those in the Long-spaced group, while the odds were more than twice as high (i.e., $e^{0.78} \approx 2.18$) for those in the Massed group when compared to learners in the Short-spaced group. When predicted probabilities were estimated, the results showed that (1) 21 % of the Long-spaced group was expected to fall into this profile; (2) 54 % of the Short-spaced group was expected to be in this profile; and (3) 72 % of Massed group was expected to be in this profile. Thus, the likelihood of being categorized into the Lower Performance Profile was highest for the Massed group, followed by the Short-spaced group and then the Long-spaced group, with the differences in odds between these groups being statistically significant.

When examining the task emotions, which is a continuous independent variable, the results illustrate the relationship between this variable and the odds of being classified into the Lower Performance Profile. Specifically, the results revealed that the odds of being categorized into the Lower Performance decreased by 63 % (i.e., $e^{-0.99} \approx 0.37$) for each one standard deviation increase in Enjoyment following the task repetition activities (i.e., at the final assessment). Aside from the Enjoyment measurement at the final assessment, no other variables demonstrated statistically significant effects on the odds of being classified into this profile. Additionally, we explored the potential interaction effect between the task repetition schedules and Enjoyment at the final assessment, but found no significant evidence. Beyond these two variables, we also assessed whether other learner factors, such as placement test, age, and gender, contributed to the odds of being classified into this profile, but no significant associations were identified.

Second, regarding the transition to the “Higher Performance Profile,” Table 9 shows the results of logistic regression models based on task repetition schedules and task emotions. For the task repetition schedules, the results revealed that the odds of being categorized into this profile were more than four times as high (i.e., $e^{1.40} \approx 4.05$) for those in the Long-spaced group, while the odds were 75 % lower (i.e., $e^{-1.37} \approx 0.25$) for those in the Massed group when compared to learners in the Short-spaced group. When predicted probabilities were estimated, the results showed that (1) 71 % of the Long-spaced group was expected to fall into this profile; (2) 38 % of the Short-spaced group was expected to be in this profile; and (3) 13 % of Massed group was expected to be in this profile. Thus, the likelihood of being

categorized into the Higher Performance Profile was highest for the Long-spaced group, followed by the Short-spaced group and then the Massed group, with the differences in odds between these groups being statistically significant.

When examining task emotions, the results indicated that the odds of being classified into the Higher Performance more than doubled (i.e., $e^{0.83} \approx 2.30$) for each one standard deviation increase in Enjoyment after the task repetition activities (i.e., at the final assessment). Aside from the Enjoyment measurement at the final assessment, no other variables showed statistically significant effects on the odds of being categorized into this profile. Similar to the findings for the Lower Performance Profile, we found no significant evidence of an interaction effect between the task repetition schedules and Enjoyment at the final assessment. Additionally, no significant associations were observed for other learner factors, such as placement test, age, and gender.

Third, concerning the profile transition to the "Unexplained Profile," Table 10 presents the results of logistic regression models. We find this analysis especially important because understanding which predictors significantly influenced learners' transition to the Unexplained Profile could provide valuable insights, given that the writing quality development of this profile remains unexplained. However, regarding the task repetition schedules, the results indicated that the odds of being classified into this profile did not significantly differ based on the type of task repetition schedules learners received during the intervention. Furthermore, the differences in odds among these groups—based on the predicted probabilities showing that (1) 8 % of the Long-spaced group was expected to fall into this profile; (2) 8 % of the Short-spaced group was expected to be in this profile; and (3) 14 % of the Massed group was expected to be in this profile—were not statistically significant. Additionally, when examining task emotions and other learner factors, such as placement test, age, and gender, the results indicated no statistically significant impact on the odds of being categorized into this profile. Taken together, we were unable to identify any factors that might have influenced learners' transition to the Unexplained Profile.

Table 7
Descriptive Statistics for Task-Related Emotions.

Variable		**Initial Assessment**			**Final Assessment**		
		Mean	**Min**	**Max**	**Mean**	**Min**	**Max**
Emotion	Anxiety (6 items)	3.35 (1.21)	1.00	5.67	3.23 (0.92)	1.50	5.67
	Enjoyment (5 items)	3.50 (1.25)	1.20	5.80	3.65 (0.94)	1.40	5.80

Note. Values in parentheses are standard deviations.

Table 8

Odds Ratios for Profile Transition to “Lower Performance Profile” Based on Task Repetition Schedules and Task Emotions.

Logistics Model	Independent Variable		Lower Performance Profile at the Final Assessment			
			B	**SE**	**OR**	**PP**
Model 1	Long-spaced		−1.49***	0.33	0.23***	0.21***
(Categorical IV)	Short-spaced		(ref.)	(ref.)	(ref.)	0.54***
	Massed		0.78*	0.32	2.18*	0.72***
Model 2	Initial Assessment	Anxiety	0.46	0.31	1.59	N/A
(Continuous IV)		Enjoyment	−0.33	0.30	0.72	
	Final Assessment	Anxiety	0.02	0.29	1.02	
		Enjoyment	−0.99**	0.29	0.37**	

Note. B represents the coefficient, SE is the standard error, OR stands for odds ratio, and PP refers to the predicted probability.

* p < .05

** p < .01,

*** p < .001,

Table 9

Odds Ratios for Profile Transition to “Higher Performance Profile” Based on Task Repetition Schedules and Task Emotions.

Logistic Regression Model	TR Types		Higher Performance Profile at the Final Assessment			
			B	**SE**	**OR**	**PP**
Model 1	Long-spaced		1.40***	0.32	4.05***	.71***
(Categorical IV)	Short-spaced		(ref.)	(ref.)	(ref.)	.38***
	Massed		−1.37***	0.38	0.25***	.13***
Model 2	Initial Assessment	Anxiety	−0.30	0.31	0.74	N/A
(Continuous IV)		Enjoyment	0.14	0.30	1.15	
	Final Assessment	Anxiety	0.19	0.31	1.21	
		Enjoyment	0.83**	0.29	2.30**	

Note. B represents the coefficient, SE is the standard error, OR stands for odds ratio, and PP refers to the predicted probability.

** p < .01

*** p < .001,

Table 10

Odds Ratios for Profile Transition to “Unexplained Profile” Based on Task Repetition Schedules and Task Emotions.

Logistic Regression Model	TR Types		Unexplained Profile at the Final Assessment			
			B	**SE**	**OR**	**PP**
Model 1	Long-spaced		0.00	0.56	1.00	0.08**
(Categorical IV)	Short-spaced		(ref.)	(ref.)	(ref.)	0.08**
	Massed		0.69	0.49	2.00	0.14***
Model 2	Initial Assessment	Anxiety	−0.48	0.54	0.62	N/A
(Continuous IV)		Enjoyment	0.52	0.54	1.68	

Logistic Regression Model	TR Types		Unexplained Profile at the Final Assessment			
			B	**SE**	**OR**	**PP**
	Final Assessment	Anxiety	−0.41	0.49	0.66	
		Enjoyment	0.42	0.46	1.52	

Note. B represents the coefficient, SE is the standard error, OR stands for odds ratio, and PP refers to the predicted probability.
** p < .01
*** p < .001,

10. Summary of the results

In summary, the present study aimed to explore how individual learners change their writing performance profiles after repeating writing tasks three times, with participants randomly assigned to one of three groups that employed different repetition schedules: Long-spaced, Short-spaced, and Massed repetition. The results indicated that most participants exhibited similar performance levels at the beginning of the intervention; however, by the end, they were classified into two distinct profiles—Lower Performance and Higher Performance, along with a smaller Unexplained Profile. Further analysis revealed that learners who received Long-spaced repetition were more likely to fall into the High-Performance Profile, while those in the Massed repetition group were more likely to be categorized as Low-Performance. Additionally, perceived enjoyment, measured after completing all writing tasks, had the strongest predictive power; specifically, greater enjoyment toward the writing tasks was associated with a higher likelihood of belonging to the High-Performance Profile.

11. Discussion

11.1. Evolving writing performance profiles of learners throughout task repetition schedules

At the initial assessment, the two-profile solution showed that most learners (93 %) fell into a Typical Profile with standardized means near zero across all CALF metrics, reflecting average writing performance likely due to limited prior exposure and early interlanguage stability (Ghyselen & De Vogelaer, 2018). The remaining 7 % formed an Unexplained Profile characterized by high accuracy but low lexical diversity, suggesting a focus on grammatical correctness at the expense of vocabulary variation (Lu & Li, 2023). This atypical distribution highlights individual writing strategies and the complexity of learner responses in early task engagement (Ellis et al., 2019).

By the final assessment, a three-profile solution emerged, illustrating the impact of task repetition schedules. The Lower Performance Profile, representing 49 % of learners and primarily associated with massed repetition, showed negative standardized means across

several CALF metrics. This may result from cognitive overload caused by the tightly clustered tasks, limiting opportunities for consolidation and reflection (Sweller et al., 2019). Motivation and engagement might also have declined due to repetition intensity, contributing to poorer performance.

In contrast, 41 % of learners in the Higher Performance Profile, mainly from short- and long-interval repetition conditions, showed positive gains in six of eight CALF metrics. This supports the spacing effect (Cepeda et al., 2006) and recent L2 writing research suggesting distributed repetition fosters improvements through increased planning, reflection, and internalization of linguistic resources (Bui & Skehan, 2018; Bygate, 2018). Rather than attributing gains to motivation, we interpret these improvements as effects of spaced repetition facilitating iterative learning and enhanced language control.

The remaining 10 % belonged to an Unexplained Profile at final assessment, exhibiting high accuracy and lexical sophistication but low fluency and complexity. This indicates a subgroup prioritizing accuracy over fluency and structural variety, reinforcing the importance of recognizing learner variability within task-based contexts (Miura, 2011).

Specifically, massed repetition participants composed most of the Lower Performance Profile (49 %) but only 11 % of the Higher Performance Profile, while short- and long-interval learners accounted for 31 % and 58 %, respectively, of the Higher Profile. This suggests spacing allows more rest, reflection, and consolidation opportunities, supporting L2 writing development (Bernacki et al., 2023). The short- and long-interval groups also showed reduced proportions in the Lower Performance Profile, further underscoring the benefits of spaced practice.

Despite these gains, differences in fluency and complexity metrics between profiles were limited, implying that task repetition alone may not suffice to improve all writing dimensions. Coupling repetition with varying task complexity may be necessary to maximize learning (Skehan, 2014). The person-centered approach here reveals diverse learner needs, suggesting future research incorporate qualitative methods to explore individual motivational and strategic differences influencing repetition outcomes (Ushioda, 2009).

Transitions between initial and final profiles revealed complex learner trajectories. Some initially in the Unexplained Profile moved to the Lower Performance Profile (22 %), while 7.5 % of Typical Profile learners shifted to the Unexplained Profile post-intervention. Nearly 44 % of those in the initial Unexplained Profile remained there, indicating idiosyncratic writing strategies resistant to change. These patterns highlight that not all learners benefit equally from repeated task exposure and underscore the need for tailored instructional approaches addressing cognitive load, motivation, and engagement (Ellis et al., 2019; Ushioda, 2009). In

short, task repetition schedules, especially spaced ones, positively shape L2 writing development, but individual learner differences significantly influence outcomes. These findings advocate for differentiated, person-centered instructional strategies in TBLT that consider learners' cognitive capacities and motivational orientations to optimize writing performance.

11.2. Task repetition schedules, emotions, and writing performance transitions

Our analysis of L2 learners' transitions across writing performance profiles reveals key interactions between task repetition schedules and emotional factors. Learners in the long-spaced repetition schedule were 77 % less likely to transition into the Lower Performance Profile compared to those in the short-spaced group. This supports prior TBLT research showing that spaced repetition promotes sustained linguistic improvement by allowing time for skill consolidation and strategic reflection (Abdi Tabari et al., 2025; Bui & Skehan, 2018; Khezrlou, 2020). Longer intervals may also aid emotional regulation, enhancing learners' readiness and confidence for repeated writing tasks, consistent with Li and Dewaele's (2024) view of learners as emotionally responsive agents adapting to task demands.

In contrast, learners in the massed (tightly sequenced) repetition condition had more than double the odds of moving into the Lower Performance Profile. This suggests that massed task schedules may impose cognitive and emotional strain, thereby limiting performance. Although cognitive load was not directly measured, these findings echo Fu and Li (2019), who highlighted the negative impact of excessive task demands on L2 writing. The high likelihood (72 %) of massed learners being classified in the Lower Performance Profile further indicates that insufficient processing time may undermine performance. These results underscore the need to design task schedules that balance cognitive and emotional demands to support L2 writing development (Güvendir & Uzun, 2023). In addition, the results imply that L2 writing assessments should carefully consider task spacing to avoid inflating cognitive load, which can distort learners' actual writing ability. Assessors interpreting results must take task sequencing into account to ensure fair and valid evaluations of learner writing performance.

Regarding transitions to the Higher Performance Profile, learners in the long-spaced group had about 16 times greater odds, and those in the short-spaced group about four times greater odds, compared to the massed group. Predicted probabilities of 71 % (long-spaced) and 38 % (short-spaced) emphasize the benefits of spaced repetition, which likely provide increased opportunities for reflection, feedback assimilation, and skill practice. This enhanced processing may boost enjoyment, which Dewaele and MacIntyre (2014) identify as critical for motivation and engagement. Emotional comfort with tasks fosters enjoyment, promoting deeper engagement and willingness to improve (Reeve, 2018).

Our analysis further revealed that each unit increase in perceived enjoyment was associated with a 63 % decrease in the odds of being classified in the Lower Performance Profile. This finding aligns with Dewaele et al. (2018), who argue that enjoyment enhances motivation and fosters positive learning experiences. While previous research focused on enjoyment's broader emotional benefits, such as reducing anxiety and promoting a positive attitude, our findings demonstrate a direct link between enjoyment and improved writing outcomes. Higher enjoyment correlates with a greater likelihood of moving into higher performance profiles, underscoring the role of emotional engagement in driving cognitive investment and task success. These results underscore the importance of fostering an emotionally supportive, enjoyable learning environment.

By contrast, anxiety showed no consistent relationship with performance profile transitions. This suggests that anxiety's influence may be moderated by other emotions such as enjoyment. This finding is consistent with Phung et al. (2021), reporting that moderate anxiety can enhance focus when tasks are perceived as achievable. Enjoyment may buffer anxiety's negative effects, helping learners to process writing tasks effectively (Abdi Tabari & Goetze, 2024; Rahimi & Zhang, 2019).

The person-centered approach in L2 motivation research provides a useful framework for interpreting these findings. While task-related emotions like enjoyment and anxiety are often treated as independent factors, this approach emphasizes how emotions interact with individual learner profiles to shape linguistic outcomes. As Ushioda (2009) states, learners bring unique motivational, cognitive, and emotional characteristics to learning. Our results suggest that enjoyment and task repetition schedules interact to enhance L2 writing outcomes, while anxiety's impact may depend on individual differences. Thus, understanding emotional effects requires considering learners' broader motivational and affective profiles, including previous experiences and emotional resilience.

Particularly, we found no significant interaction between task repetition schedules and emotional states, indicating that these factors independently influence writing performance without amplifying or diminishing each other's effects. This underscores the importance of simultaneously optimizing task schedules and fostering positive emotions in instructional design. Given that emotional fluctuations during repeated tasks can produce varied outcomes, well-designed tasks should minimize excessive anxiety while promoting enjoyment. Additionally, learner characteristics such as age, gender, and placement scores showed no significant association with performance profile transitions, underscoring the central role of task repetition schedules and emotions. These findings suggest that writing assessments should independently consider both task sequencing and emotional factors in their design to ensure

accurate measurement of L2 writing ability. Assessors interpreting writing performance must also account for these separate influences to avoid confounding actual writing skills with situational factors.

Overall, our findings support instructional strategies that combine long-spaced repetition with efforts to enhance enjoyment and engagement. Such approaches can significantly improve L2 writing performance and learner experience in TBLT contexts. By applying a person-centered perspective, L2 writing instructors can better appreciate learner diversity and tailor learning environments to support cognitive, emotional, and motivational development. These results suggest that L2 writing assessments should incorporate task spacing to minimize cognitive and emotional overload, ensuring more accurate measurement of L2 writing ability.

12. Conclusion: implications, limitations, and directions

The findings of this study contribute to our understanding of task-based L2 writing instruction by demonstrating the effects of the spacing effect and employing a person-centered approach through latent transitional analysis (LTA). The positive results of spaced task repetition highlight the potential of distributing practice over time to enhance writing performance, aligning with the principles of study-phase retrieval theory. However, as the study focuses solely on writing performance during task engagement over three weeks, without a posttest component, it does not fully address the long-term retention implications of study-phase retrieval theory. Similarly, while the variation in learners' writing profiles offers insights into task challenge and engagement, the desirable difficulty framework remains only partially addressed. This framework typically requires evaluating both task engagement and subsequent learning outcomes, which were beyond the scope of this research. The findings suggest that while challenging tasks can promote engagement, the calibration of task difficulty must be explored further to optimize learning outcomes, particularly in relation to long-term retention and cognitive processing. These limitations highlight the need for future research to more comprehensively investigate the interplay between task design, desirable difficulty, and study-phase retrieval in L2 writing contexts.

To implement these findings effectively, L2 writing teachers should prioritize spaced task repetition schedules over massed practices. The positive results seen in learners assigned to short- and long-interval repetition schedules suggest that providing time for cognitive processing can significantly enhance writing performance. Teachers are encouraged to design writing tasks that incorporate reflection periods, fostering deeper engagement with the material. Given the diversity of performance profiles among learners, a differentiated instructional approach is essential. Recognizing that students respond differently to tasks, L2 writing teachers should offer tailored feedback and adjust task complexities to meet the unique

needs of individual learners, particularly those who may face challenges in maintaining cognitive load in the massed condition. While the massed repetition condition may increase cognitive fatigue, further research is needed to directly measure and confirm cognitive overload in this context. Additionally, the significant relationship between perceived enjoyment and performance outcomes underscores the importance of fostering a supportive learning environment that cultivates enjoyment in writing tasks. Teachers can enhance emotional engagement by integrating strategies such as collaborative writing projects and opportunities for peer feedback, which can help mitigate anxiety and promote motivation.

However, this study has several limitations. First, the small sample size may restrict the generalizability of the findings, as learner variability can affect the outcomes of task-based instruction. Additionally, the study focused on specific CALF metrics, which may not encompass all dimensions of L2 writing performance, overlooking qualitative aspects of L2 writing development. The reliance on self-reported measures of emotional states, such as anxiety and enjoyment, may introduce bias, as these perceptions may not accurately reflect learners' true emotional experiences during task performance. Finally, the duration of the intervention may not have been sufficient to capture the long-term effects of task repetition schedules on L2 writing performance.

Thus, future research should investigate the impact of varying task complexities alongside different repetition schedules to further understand their interplay in L2 writing TBLT contexts. Exploring the role of individual learner characteristics, such as prior experience and cognitive processing styles, could yield deeper insights into the factors influencing writing performance. Longitudinal studies examining the long-term effects of spaced practice on writing skills and emotional dynamics would also be beneficial. Additionally, qualitative research exploring learners' experiences and perceptions during task-based writing instruction could enrich the understanding of how emotional factors interact with instructional design. Finally, comparative studies across diverse educational contexts and language backgrounds could illuminate the broader applicability of the findings and contribute to the development of more effective task-based writing pedagogies.

CRediT authorship contribution statement

Mahmoud Abdi Tabari: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization. **Hansol Lee:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation.

Data Availability

The data that has been used is confidential and cannot be shared.

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Mahmoud Abdi Tabari is Teaching Assistant Professor of TESOL and Writing Studies at the University of Nevada, Reno (UNR). His research focuses on second language acquisition, second language writing, and task-based language teaching. He has published his work in various top-tier academic journals, including *Applied Linguistics, Modern Language Journal, Journal of Second Language Writing, Language Teaching Research, System, Reading and Writing, Assessing Writing, and Journal of English for Academic Purposes*.

Hansol Lee is a Professor of English at Korea Military Academy, South Korea. He received his PhD degree in Education from the University of California, Irvine. His research interests include methods and statistics, education, and literacy. His work has been published in *Review of Educational Research, Education Research Review, Child Development, Applied Measurement in Education, Applied Linguistics, Language Learning, The Modern Language Journal, Language Learning & Technology, Journal of Language and Social Psychology, Annual Review of Applied Linguistics*, among others.

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3. Exact task repetition refers to the repeated performance of the *same* task with the *same input, structure, and goal*, allowing learners to apply similar strategies and knowledge across attempts (Bygate, 2018).

4. Procedural repetition involves repeating tasks that share *similar procedures or formats* but differ in specific content or topics. This type of repetition engages learners in familiar task routines while requiring them to process new input (Lázaro-Ibarrola & Hidalgo, 2017).
5. This study refers to scheduling patterns as the timing and distribution of task repetition across sessions (e.g., massed vs. spaced) (Abdi Tabari et al., 2025).
6. Spaced practice involves distributing repetitions over longer intervals, which has been shown to enhance long-term retention and performance (Suzuki & DeKeyser, 2017).